

4. What is Machine Learning

What is Machine Learning?



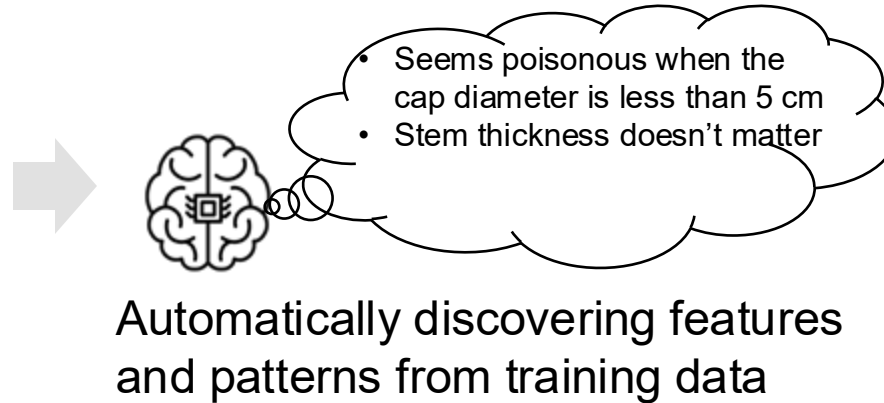
A technology in which computers automatically learn features and patterns from data, rather than being explicitly given rules or laws by humans. As a result of this training, computers can predict desired outcomes even for unknown data.

Loading data into a computer

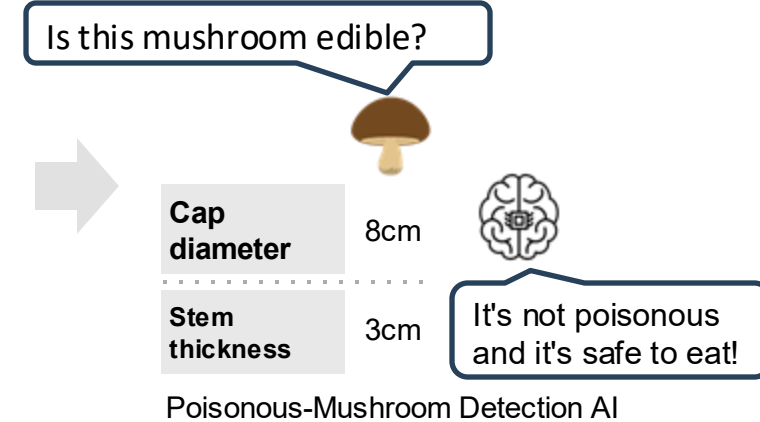
Example: Data of mushrooms

Cap diameter				
	10cm	5cm	7cm	4cm
Stem thickness	1cm	2cm	5cm	2cm
Poisonous?	No	Yes	No	Yes

Learning from data



Prediction



Machine learning is also used in the following areas:

- Detecting spam emails
- Forecasting weather
- Recommend products on e-commerce sites

Types of Machine Learning

Machine learning can be separated to three categories: supervised learning, unsupervised learning, and reinforcement learning.

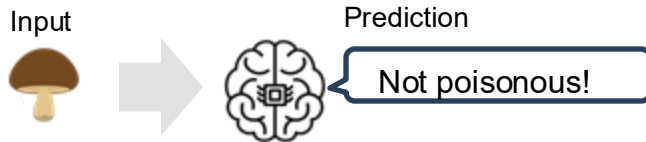
Supervised Learning

- A correct answer is provided.
- The model learns the relationship between input and answer, then predicts the answer for new input.

e.g. Mushroom data

			
Cap diameter	10cm	5cm	Inputs (=Features)
Stem thickness	1cm	2cm	
Poisonous?	No	Yes	

After Training



Unsupervised Learning

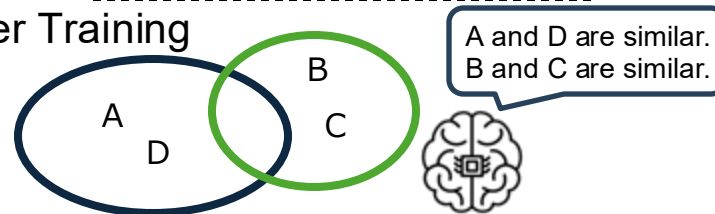
- A correct answer is NOT provided.
- The model discovers patterns or relationships within the input data.

e.g. Test scores

	A	B	C	D
				
Math	90	75	60	70
Science	80	60	50	90
Social St	65	95	80	70
English	80	70	90	70

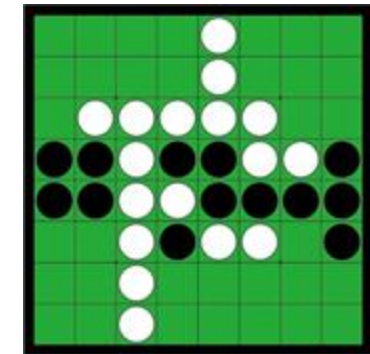
No answer data

After Training

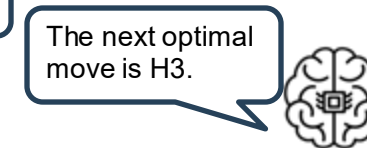


Reinforcement Learning

- The model receives different rewards depending on the actions it chooses and learns actions that maximize total reward.

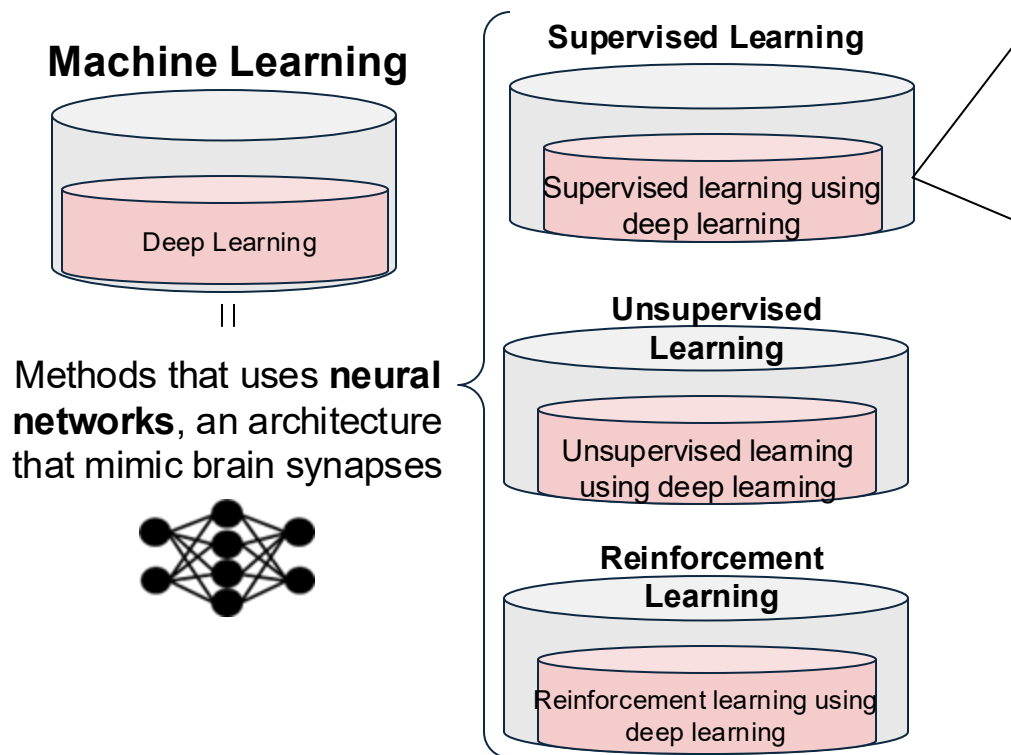


After Training



Deep Learning

The technique using a multi-layered “neural network” model is called deep learning, broadly referring to any learning scheme where the model repeatedly performs trial-and-error, automatically adjusting features.



Ordinary Machine Learning

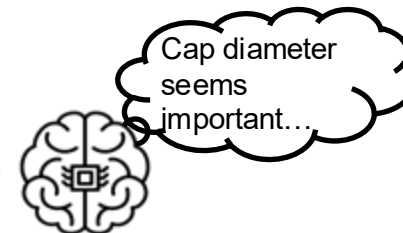
Cap Diameter	10cm	5cm	7cm	4cm
Stem Thickness	1cm	2cm	5cm	2cm
Poisonous?	No	Yes	No	Yes

In ordinary machine learning workflows, humans must decide which measurements to use as “features.”

Deep Learning

Poisonous?	No	Yes	No	Yes
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Automatically discovers and adjusts features by trial-and-error.
Applied to difficult tasks such as image data.



Quiz - Check Your Understanding



Think of words that go in the blank spaces.

[?]

A method where the model automatically discovers actions that will earn more rewards by assigning different rewards based on the selected actions.

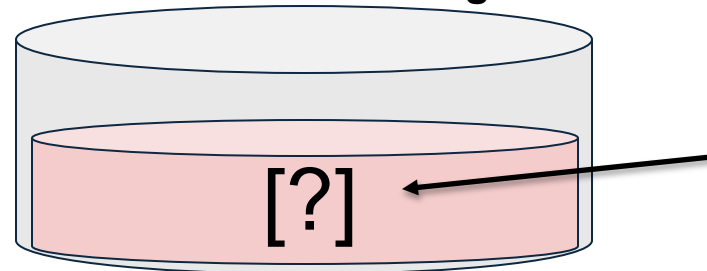
[?]

A method where labeled data (i.e., answers) is available, and the model learns the relationship between inputs and outputs in order to make predictions from input data.

[?]

A method where there is no labeled data, and the model finds relationships by identifying patterns in the input data.

Machine Learning



Neural network based analyses that mimic brain synapses

Quiz - Check Your Understanding (Answer)

Think of words that go in the blank spaces.

[Reinforcement learning]

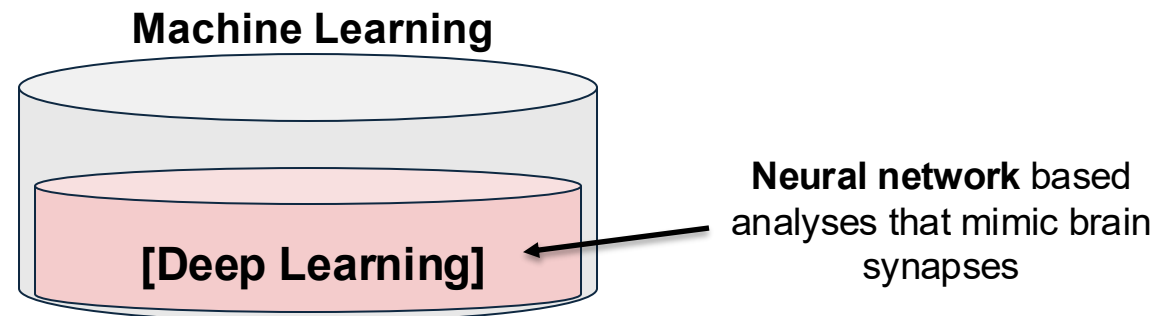
A method where the model automatically discovers actions that will earn more rewards by assigning different rewards based on the selected actions.

[Supervised learning]

A method where labeled data (i.e., answers) is available, and the model learns the relationship between inputs and outputs in order to make predictions from input data.

[Unsupervised learning]

A method where there is no labeled data, and the model finds relationships by identifying patterns in the input data.



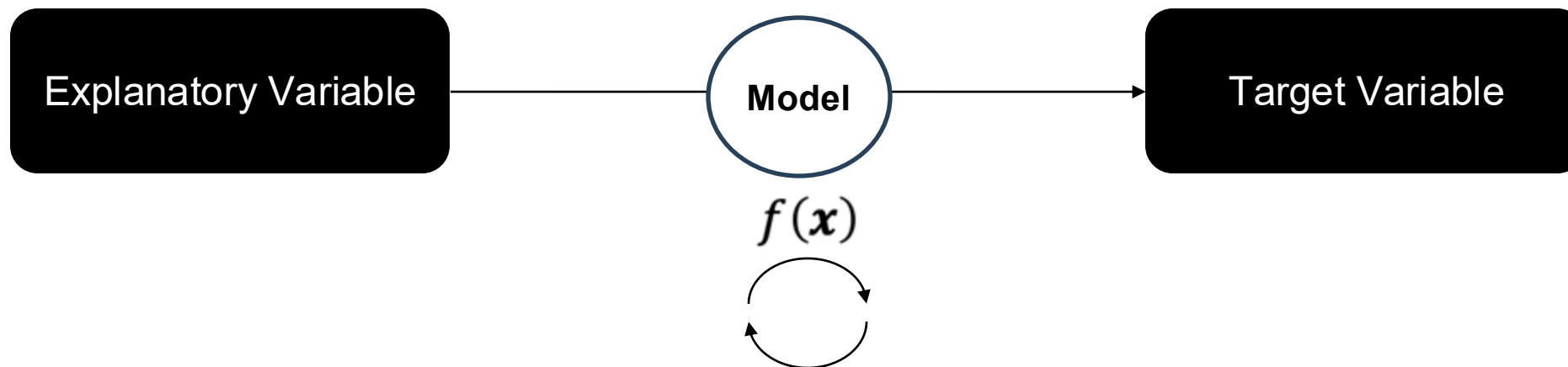
Supervised Learning

Supervised Learning



Supervised learning trains a model to predict a target variable from explanatory variable (features) and adjusts the model so that its outputs closely match the targets.

Concept of Supervised Learning



The model is repeatedly adjusted until its output gets closer to the target variable.

Main Types of Supervised Learning

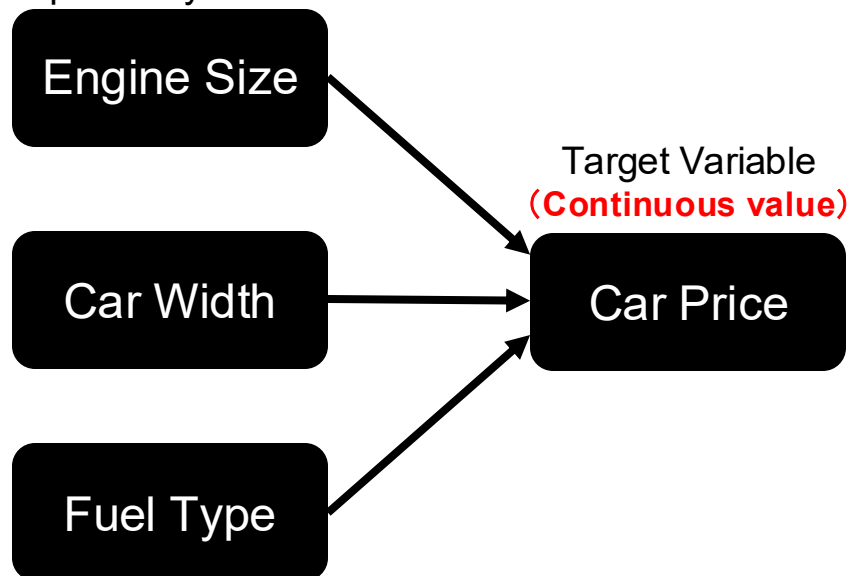


The method is generally divided into two types: regression and classification.

- Regression: If the target variable is continuous values
- Classification: If the target variable is discrete value and becomes a category (i.e. whether a mushroom is “poisonous” or “safe”)

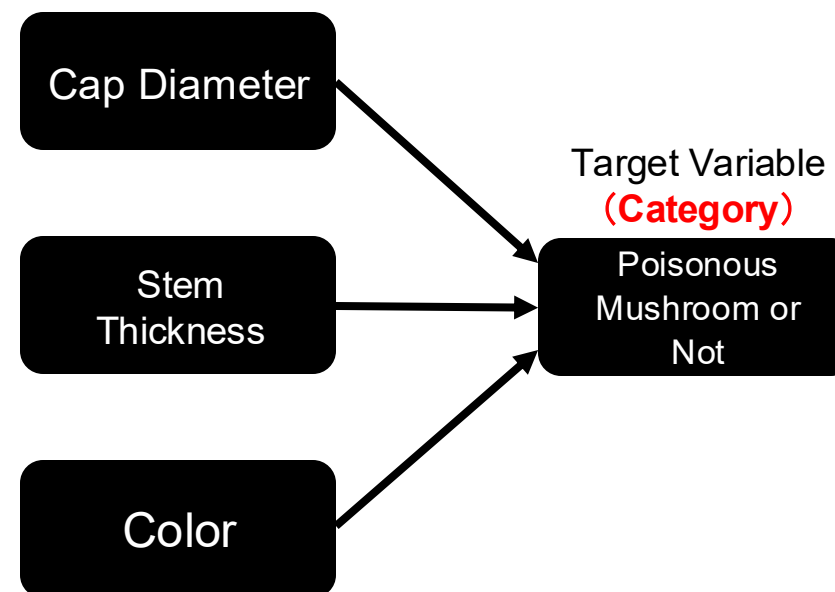
Regression

Explanatory Variable



Classification

Explanatory Variable



Supervised Learning (Regression)



The most basic model is linear regression, which represents the target variable as a weighted sum of several explanatory variables plus a bias term. By adjusting the weights, we can build an optimal predictive model.

Linear Regression Formula

$$f(x) = \underbrace{w_1 x_1}_{\text{weight} \times \text{explanatory variable}} + w_2 x_2 + w_3 x_3 + \cdots + w_n x_n + \underbrace{b}_{\text{bias}}$$

The **target variable** ($f(x)$) is expressed as a weighted sum of multiple **explanatory variable**(x), plus a **bias** term (b).

The weight indicates how much influence each explanatory variable has on the target variable.

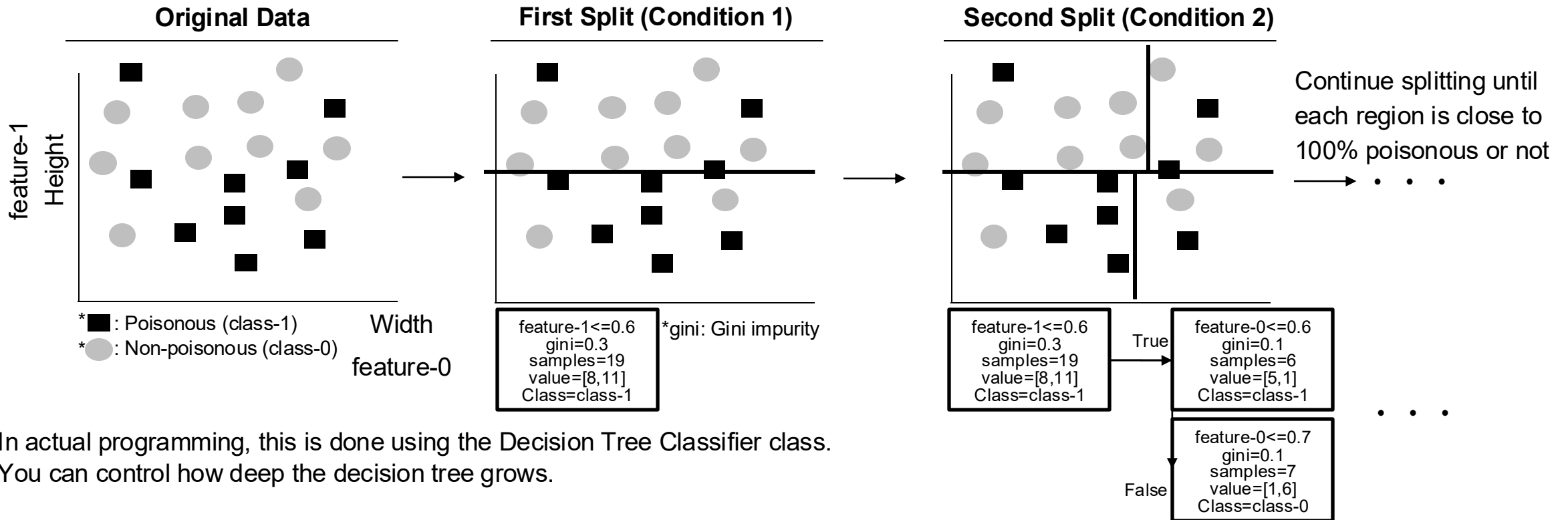
The bias is the value when all explanatory variables are 0. It serves as the baseline of the model.

By using this equation, we adjust the weights so that the difference between the predicted results and the correct data becomes small, and construct the optimal prediction model.

Supervised Learning (Classification)(1/2)

A common classification method is the Decision Tree, which classifies data by recursively branching based on attribute conditions.

Visualization of Decision Tree Analysis



*In actual programming, this is done using the Decision Tree Classifier class.

*You can control how deep the decision tree grows.

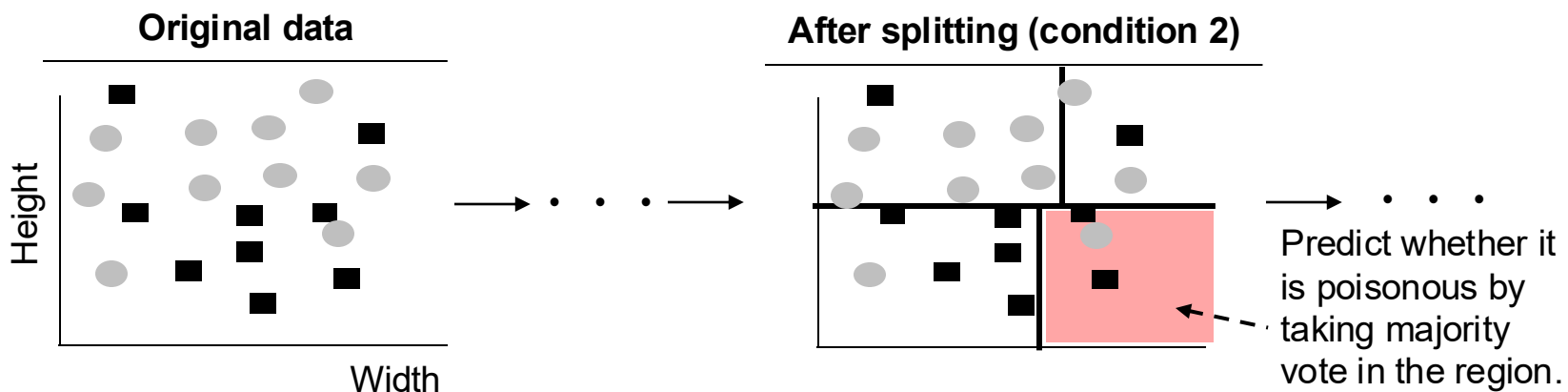
(cf.) Regression Trees and Classification Trees

There are two types of decision trees.

- **Classification tree**: when the target variable is a category
- **Regression tree**: when the target variable is a numerical value

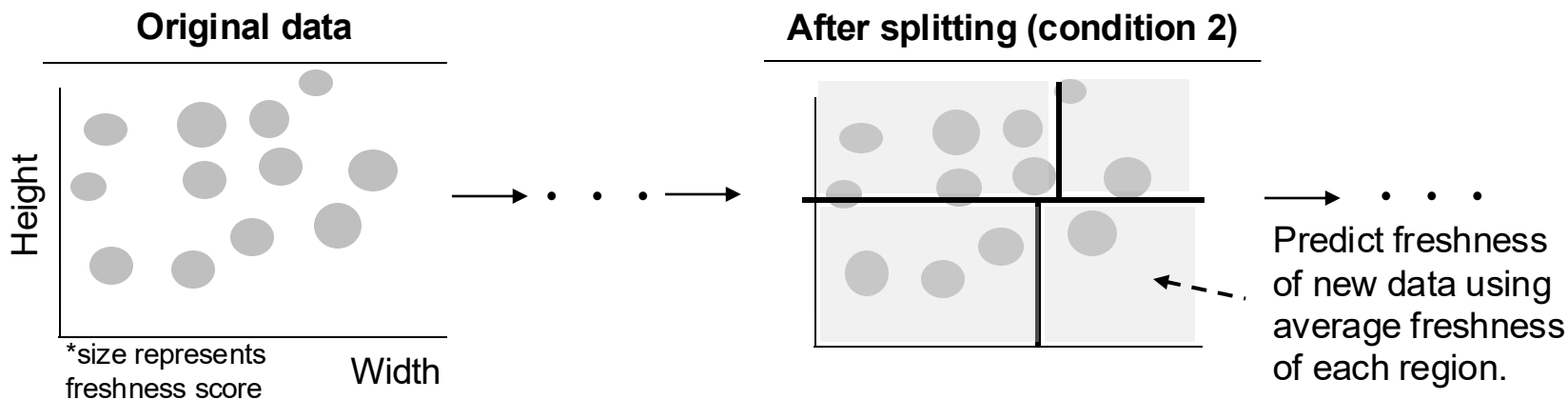
Classification Tree

Target :
Poisonous or
Not
(Category)



Regression Tree

Target :
Freshness
score
(Continuous
value)



Supervised Learning (Classification)(2/2)

Another classification method is logistic regression. This model transforms any input values into a value between 0 and 1, and is used to predict categorical data such as Yes or No.

Logistic Regression Formula

$$f(x) = \frac{1}{1 + \exp(-(w_1x_1 + w_2x_2 + \dots + w_nx_n + b))}$$

This is essentially a linear regression formula inside a sigmoid function.

By inserting the linear regression formula into the sigmoid function, you can obtain values between 0 and 1, which allows for classification into Yes or No.

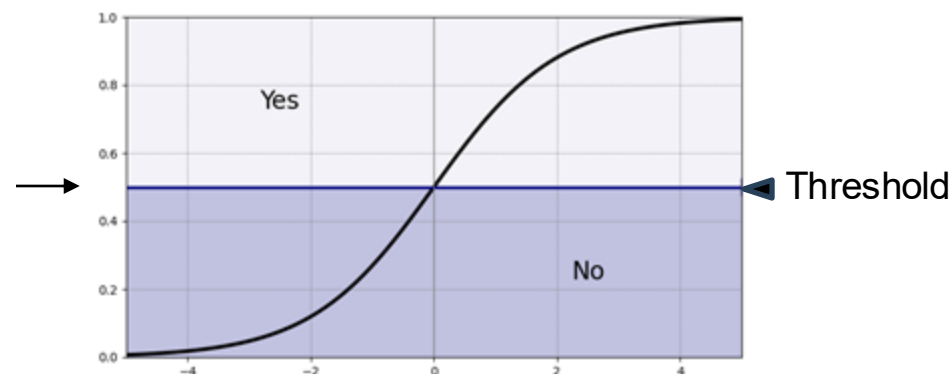
*It may also be expressed like this:

$$\text{sigmoid}(w_1x_1 + w_2x_2 + w_3x_3 + \dots + w_nx_n + b)$$

*The term “logistic” is used for models representing growth or decrease.

*For now, it's okay to think of the sigmoid function simply as a function that converts any value into something between 0 and 1

Computation Result



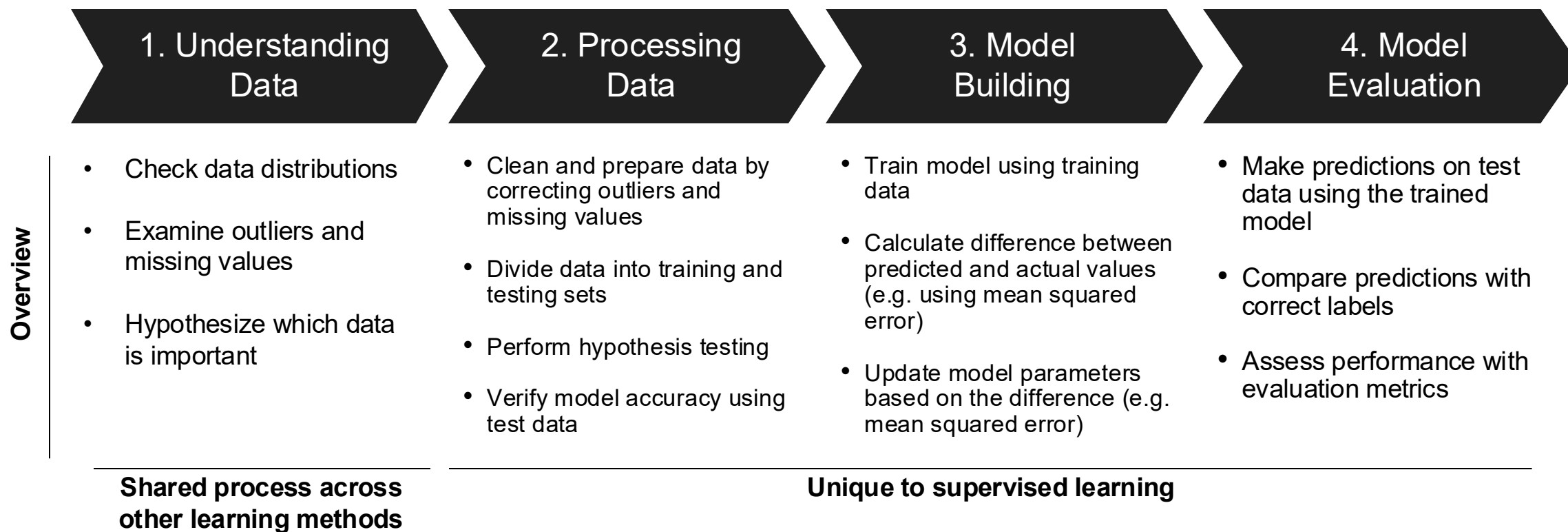
When the output value falls between 0 and 1, you can classify the result into categories such as Yes or No by setting a **threshold**.

Supervised Learning Workflow



The overall workflow is largely similar to Data Science Workflow (see Section 1). However, Step 2 (Processing Data) to 4 (Model Evaluation) follow a distinct process.

Supervised Learning Analysis Process

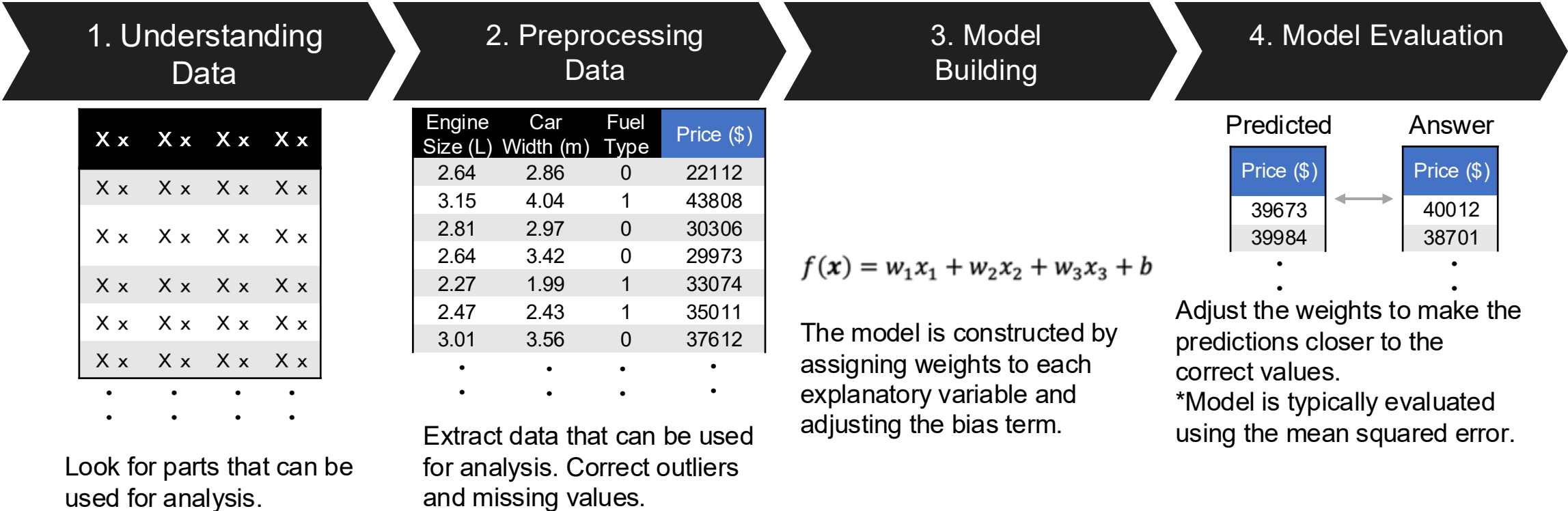


Supervised Learning (Regression Case)



Steps like Step 2: Processing Data and Step 3: Model Building —particularly how to adjust feature weights—are especially important for achieving good results.

Linear Regression Analysis Example

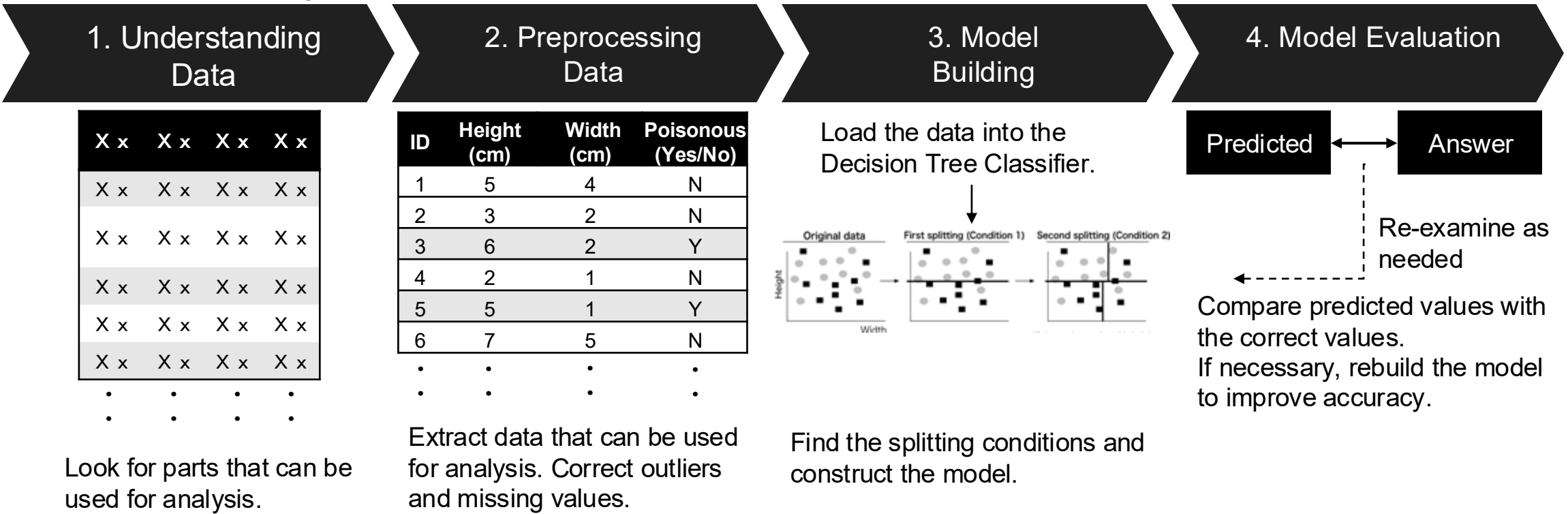


Supervised Learning (Classification Case)



Even in analyses using methods like decision trees, Step 2: Preprocessing Data and Step 3: Model Building are critical for achieving good results.

Decision Tree Analysis Example



Quiz - Check Your Understanding



Think of words that go in the blank spaces related to Supervised Learning.

[?]

A **regression** method that expresses the target variable as the weighted sum and bias of multiple explanatory variables.

[?]

A **classification** method that uses a model to map values to a range between 0 and 1, and predicts qualitative data such as Yes or No.

[?]

A method that repeatedly splits data based on attribute conditions to classify them. Can be used for both **regression and classification** tasks.

Quiz - Check Your Understanding (Answer)



Think of words that go in the blank spaces related to Supervised Learning.

[Linear Regression]

A **regression** method that expresses the target variable as the weighted sum and bias of multiple explanatory variables.

[Logistic Regression]

A **classification** method that uses a model to map values to a range between 0 and 1, and predicts qualitative data such as Yes or No.

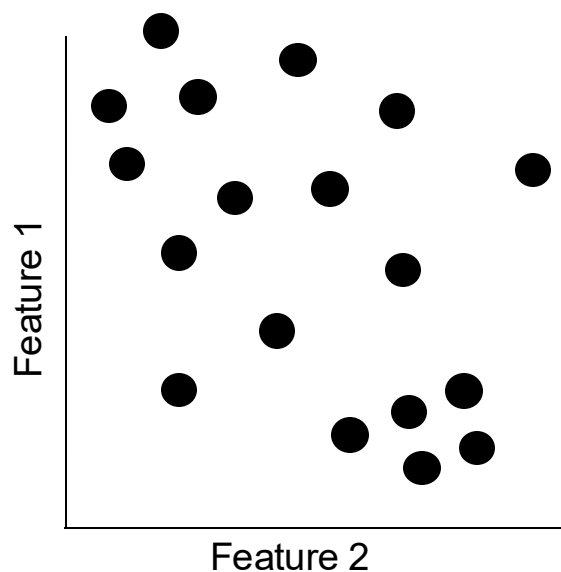
[Decision Tree]

A method that repeatedly splits data based on attribute conditions to classify them. Can be used for both **regression and classification** tasks.

Unsupervised Learning

A method for discovering hidden patterns and insights in input data. Since it focuses solely on the input data, it is used when no target variable can be defined.

Data to Be Analyzed

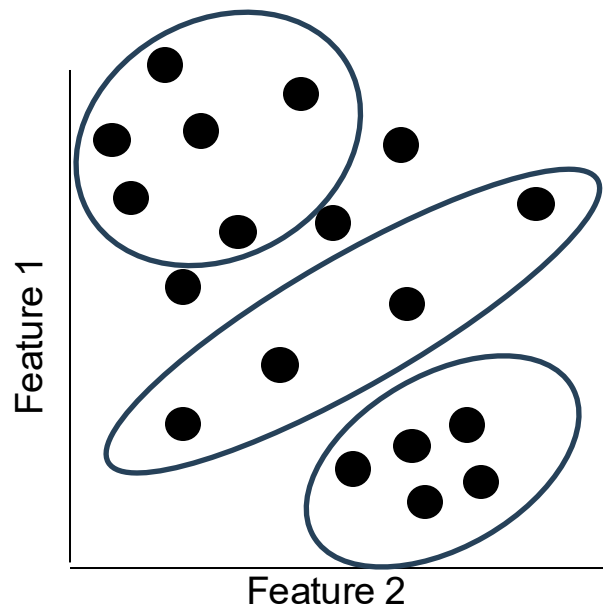


Unclear what insights can be given from this unorganized data.

Analyzing whether the data itself shows any trends, even though the goal is unclear.



Results of Unsupervised Learning



Understanding trends in the data makes it easier to draw insights.

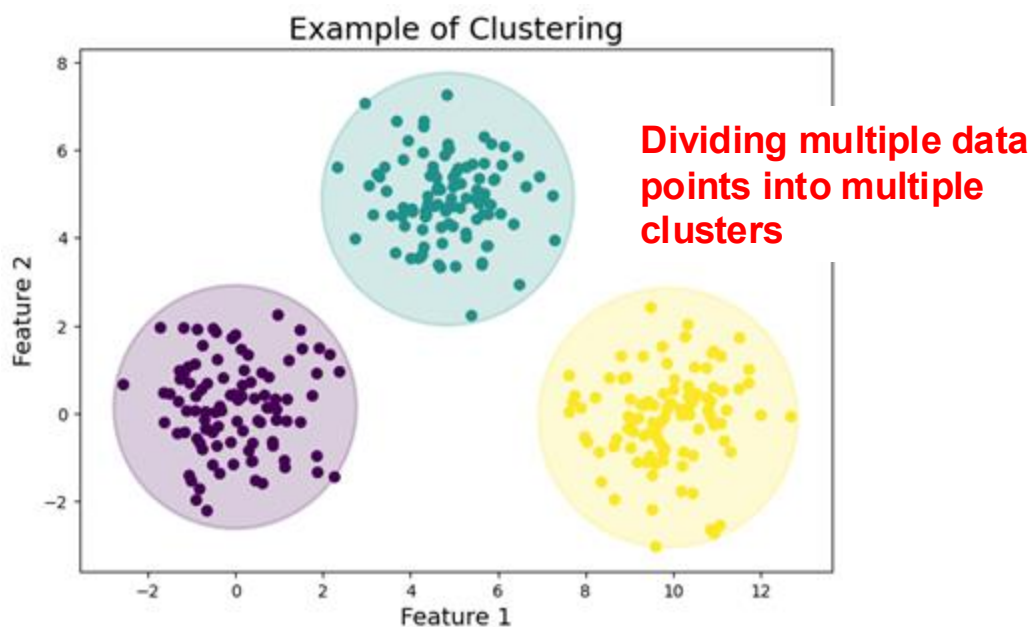
Unsupervised Learning

Examples of Unsupervised Learning

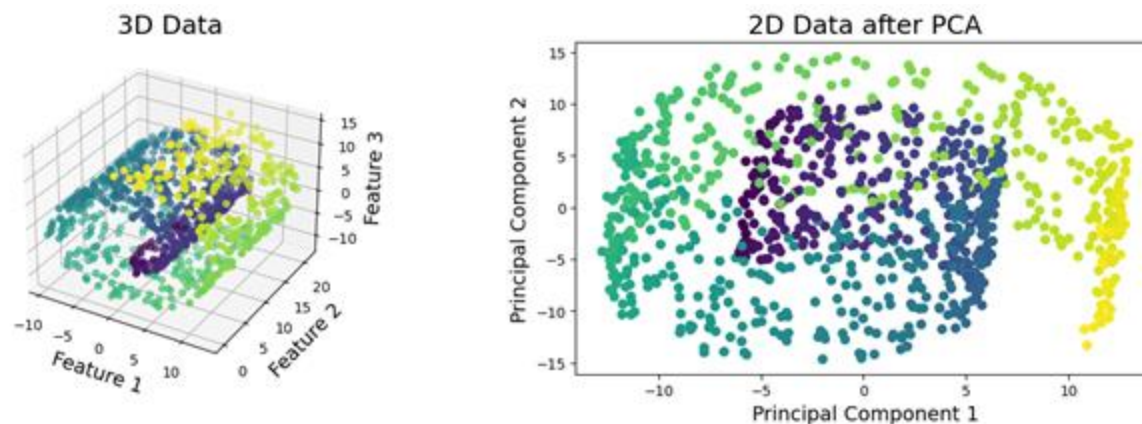


Major unsupervised learning techniques include clustering and dimensionality reduction methods such as Principal Component Analysis (PCA).

Clustering



Dimensionality Reduction



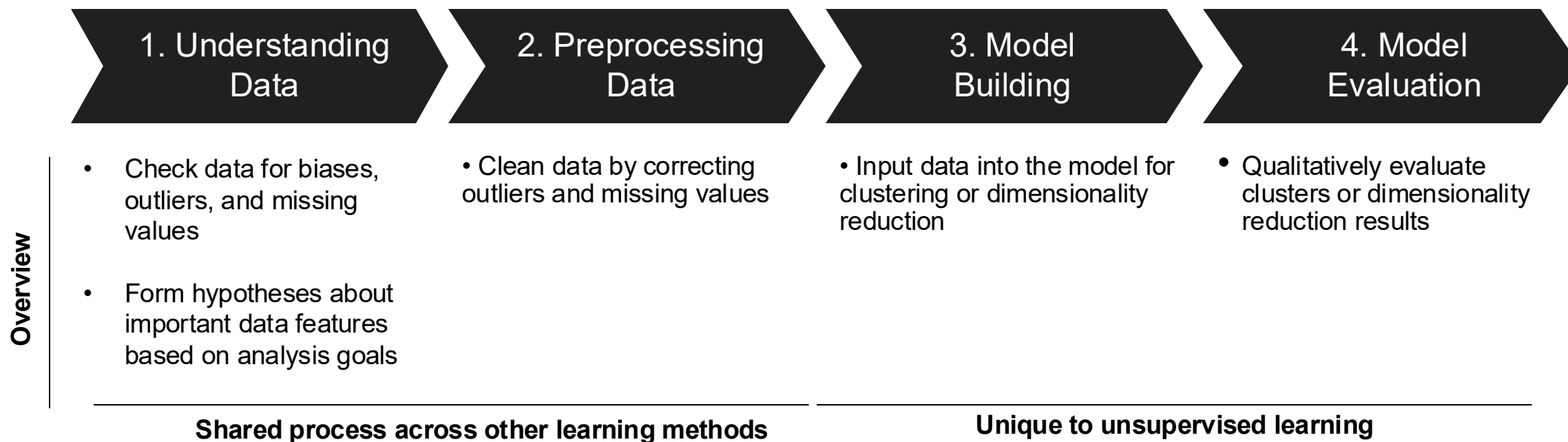
Compressing 3D data into 2D data

Unsupervised Learning Workflow



The process consists of four steps (understanding data, processing data, model building and model evaluation), just like the Data Science Workflow (see Section 1).

Unsupervised Learning Analysis Process



Unsupervised Learning (Clustering Case)



Instead of spending time on 1. understanding data, make hypotheses from clusters as a result of 3. model building.

Clustering Analysis Example

1. Understanding Data

X x	X x	X x	X x
X x	X x	X x	X x
X x	X x	X x	X x
X x	X x	X x	X x
X x	X x	X x	X x
X x	X x	X x	X x
:	:	:	:

Look for parts that can be used for analysis.

2. Preprocessing Data

Store	Cosmetic	Clothing	Food
StoreA	15	30	40
StoreB	10	10	100
StoreC	90	15	25
StoreD	30	25	50
StoreE	100	10	15
:	:	:	:

Extract data that can be used for analysis. Correct outliers and missing values.

3. Model Building

Use a model like K-means* to group data based on how close or far they are from each other.

*A method that sets a center point and clusters based on distance from that center

4. Model Evaluation

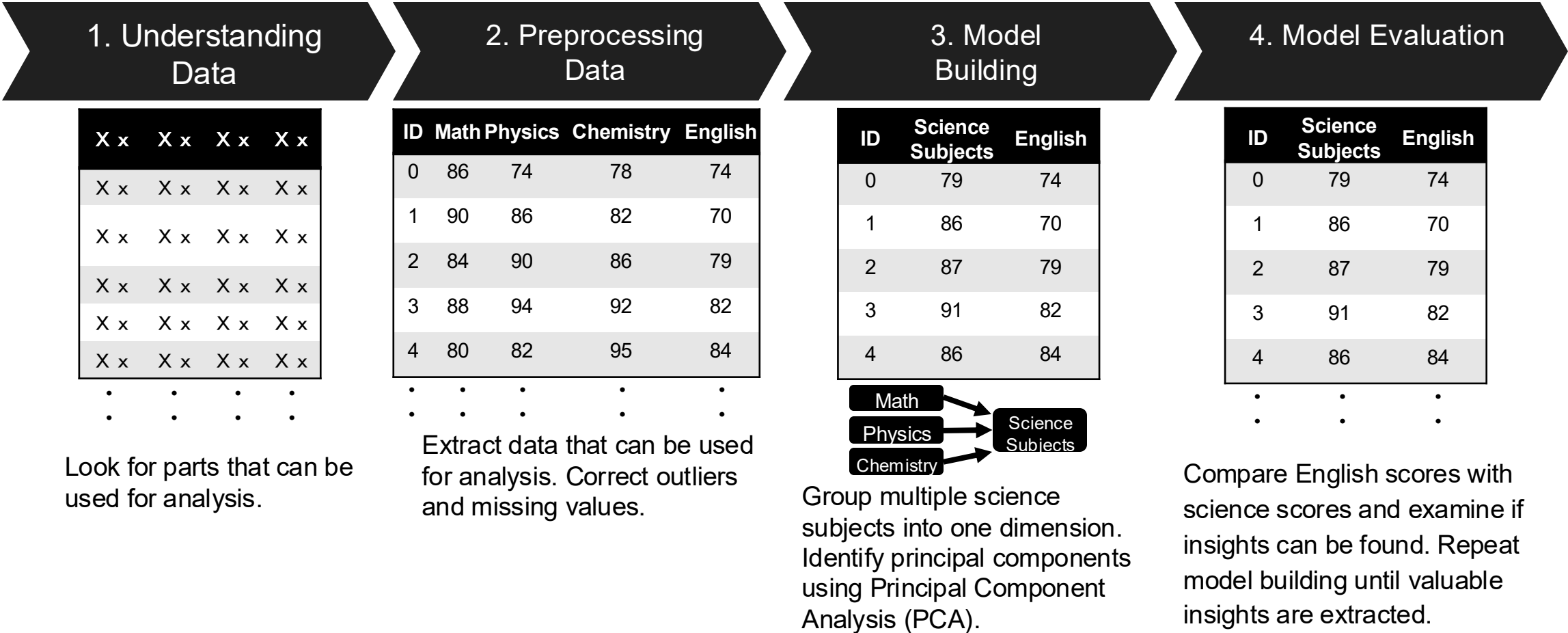
Repeat clustering until you find good clusters that provide insights.

Unsupervised Learning (Dimensionality Reduction Case)



Dimensionality reduction is often used when there are many types of data, and it is difficult to clarify the relationship. It helps make analysis simpler.

Dimensionality Reduction Analysis Example



Quiz - Check Your Understanding



Think of words that go in the blank spaces related to Unsupervised Learning.

[?]

A method of dividing a large amount of data into several similar groups.

[?]

A method of compressing high-dimensional data into a lower-dimensional space without losing important information.

Quiz - Check Your Understanding (Answer)



Think of words that go in the blank spaces related to Unsupervised Learning.

[Clustering]

A method of dividing a large amount of data into several similar groups.

[Dimensionality Reduction]

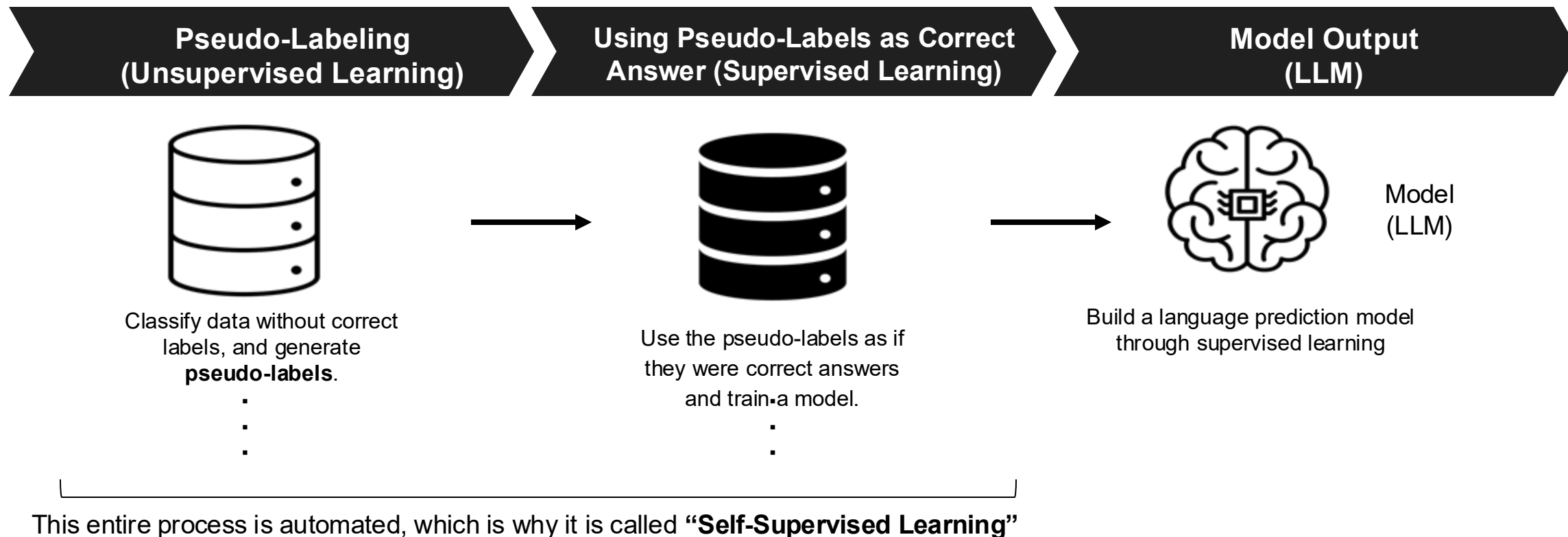
A method of compressing high-dimensional data into a lower-dimensional space without losing important information.

Application of Machine Learning in LLM

Large Language Model (LLM)



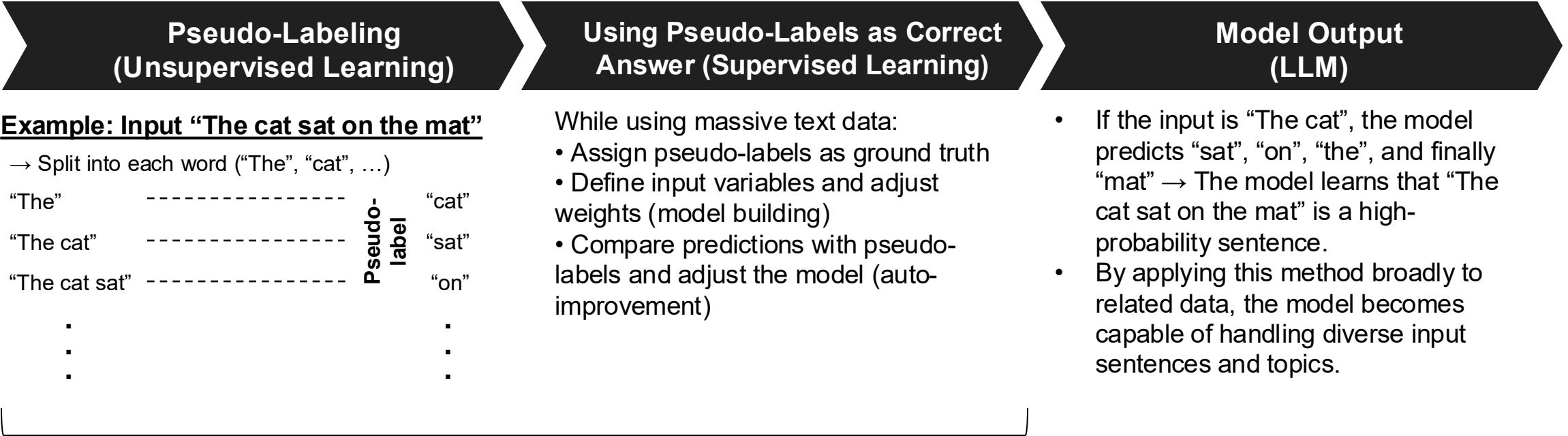
A method called self-supervised learning is used in the training of large language models or LLMs, incorporating concepts from both supervised and unsupervised learning.



Learning Method of Large Language Models (LLMs)



LLMs are built using **Next Token Prediction** (a method that estimates the probability of the next word appearing), a type of **self-supervised learning**. As a result, the model becomes capable of predicting the next word based on part of the input.



This entire learning process is called “Next Token Prediction”

Quiz - Check Your Understanding



Think of words that fit in the blank spaces.

**Pseudo-Labeling
(Unsupervised Learning)**

**Using Pseudo-Labels as Ground
Truth (Supervised Learning)**

**Model Output
(LLM)**



Classify data without correct labels, and generate **pseudo-labels**.



Use the pseudo-labels as if they were correct answers and train a model.



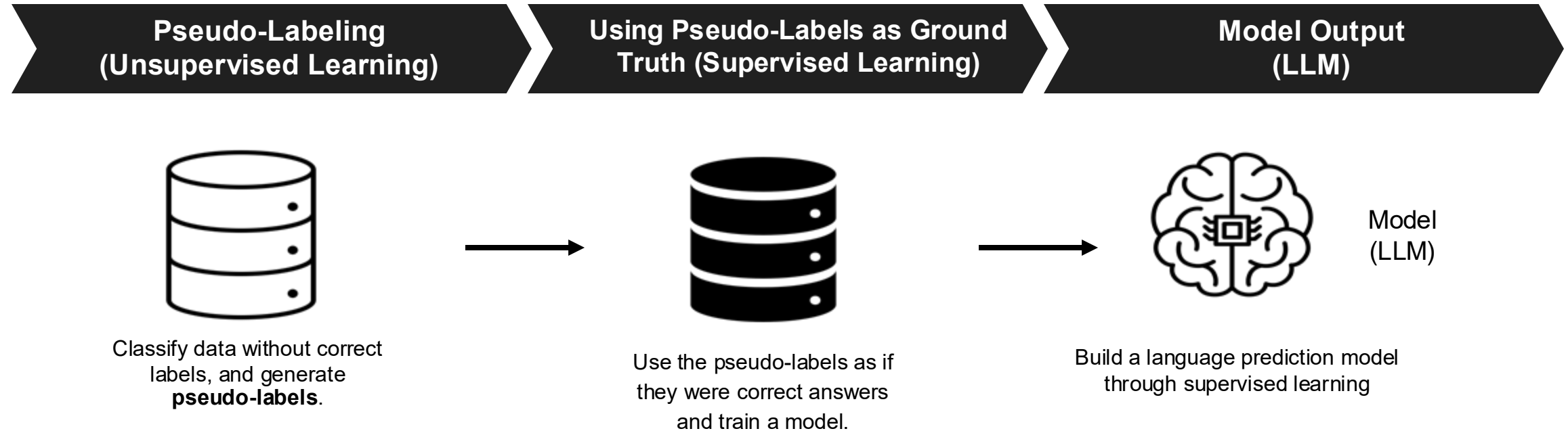
Model
(LLM)

Build a language prediction model through supervised learning

LLMs are trained through [], which incorporates concepts from both supervised learning and unsupervised learning

Quiz - Check Your Understanding (Answer)

Think of words that fit in the blank spaces.



LLMs are trained through [**self-supervised learning**], which incorporates concepts from both supervised learning and unsupervised learning